P&L Explain / Exception Dashboard for a Simulated Spot FX Trading Book

Github repo

Objective:

This project simulates a middle-office analyst supporting a small spot FX trading book. The objective was to calculate expected daily P&L, compare it against a separate reported P&L source, identify material variances, classify likely root causes, and present the results in a dashboard-style reporting layer.

Project scope:

- 45 business days
- 3 FX pairs: EURUSD, GBPUSD, AUDUSD
- Daily prices, trades, fees, and derived positions
- Materiality threshold: absolute P&L variance > $150
- Exception types: missing trade, duplicate booking, stale price, wrong position, incorrect fee

Workflow:

Clean Data → Expected P&L Engine → Reported P&L with Planted Breaks → Exception Detection → Dashboard + Investigation Notes

Outputs / Results:

- Built a Python-based expected P&L engine using positions, trades, prices, and fees
- Calculated daily realized P&L, unrealized change P&L, and fee-adjusted expected total P&L
- Injected 10 controlled exception scenarios across 5 break types
- Developed a rule-based exception engine to classify likely root causes
- Achieved 100% match against planted break scenarios in validation testing
- Produced dashboard summaries showing daily P&L, variance trends, exception counts, and severity

Tools used: Python, pandas, numpy, matplotlib, CSV-based reporting workflow

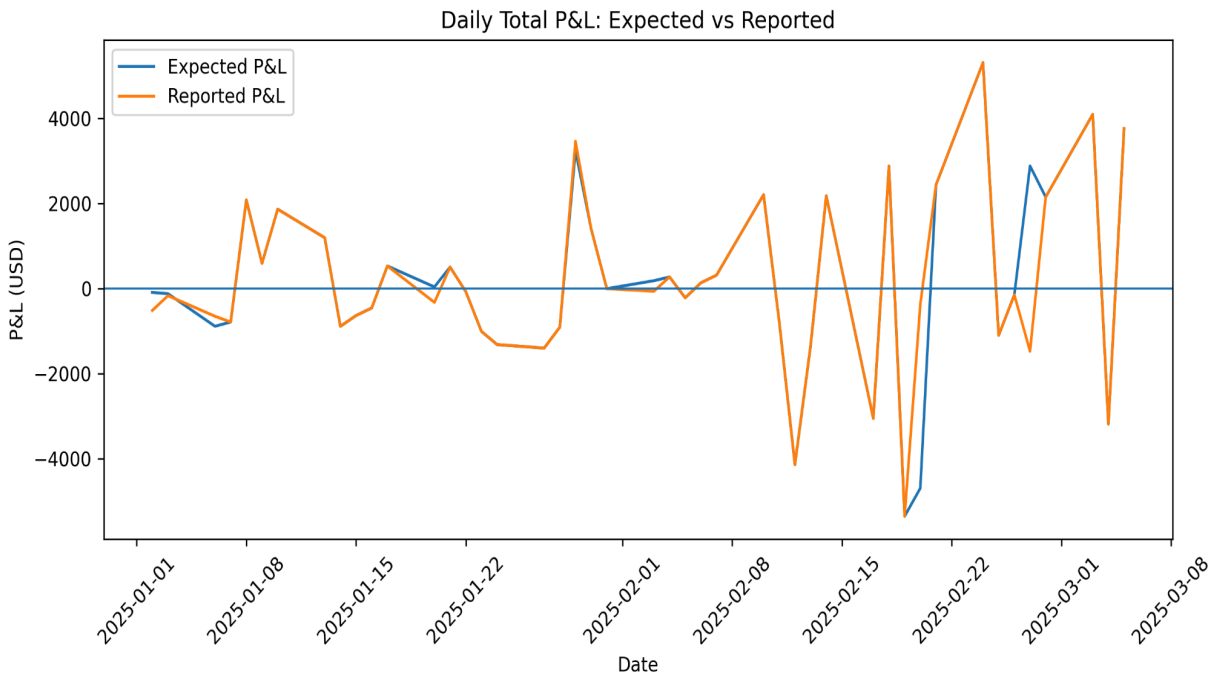
Controls:

- trade ID uniqueness checks
- fee completeness checks
- position continuity validation
- position roll-forward validation
- price completeness checks
- material variance flagging and severity ranking

Relevance:

This project was designed to mirror middle-office / trading operations responsibilities including daily P&L explain, exception handling, break triage, and control-focused reporting.

Visuals:



Row Labels	Count of likely cause	Average of abs PNL variance	Max of abs PNL variance	Average of age business days
Duplicate booking	2	305.355	362.86	27
Incorrect fee	2	212.5	250	44
Missing trade	2	229.65	235.45	33.5
Stale price	2	4339.625	4356.75	6.5
Wrong position	2	197.625	220.5	43
Grand Total	**10**	**1056.951**	**4356.75**	**30.8**